

4.4.4 Variation and evolution

An understanding of the interplay between evidence and theory in the development of scientific thinking about evolution by natural selection and the classification of living organisms has enabled scientists to develop technologies to make agriculture more productive by means of selective breeding and genetic engineering. These technologies raise ethical issues.

4.4.4.1 Mutations

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
State that there is usually extensive genetic variation within a population of a species. Recall that all variants arise from mutations, and that most have no effect on the phenotype, some influence the phenotype and a very few determine the phenotype.	Mutations are changes in DNA molecules that may affect genes. Mutation of a gene can alter the proteins that it contains the code for, or even prevent the protein being produced in cells. Mutations can happen when DNA is copied during cell division or when cells are affected by environmental factors such as ionising radiation.	

4.4.4.2 Evolution through natural selection

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe evolution as a change in the inherited characteristics of a population over time through a process of natural selection which may result in the formation of new species. Explain how evolution occurs through natural selection of variants that give rise to phenotypes best suited to their environment.	The theory of evolution by natural selection explains the evolution of all species of living things from simple life forms that first developed more than three billion years ago. If two populations of one species become isolated geographically or environmentally they may evolve in different ways to suit different conditions. If they become so different that they can no longer interbreed to produce fertile offspring they have formed two new species.	WS 1.2 Use the theory of evolution by natural selection in an explanation.

4.4.4.3 Evidence for evolution

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the evidence for evolution, including fossils and antibiotic resistance in bacteria.	Evidence for evolution comes from the study of fossils that show how much or how little different organisms have changed as life developed on Earth. Evolution of bacteria can be observed happening in a much shorter time because they reproduce so fast. Bacteria that cause disease evolve by natural selection when exposed to antibiotics; this gives rise to a resistant strain.	MS 2c, 4a Extract and interpret information from charts, graphs and tables.

4.4.4.4 Identification and classification of living things

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe the impact of developments in biology on classification systems.	In studies of evolution it is essential to be able to identify and classify living things. Traditionally living things have been classified into groups depending on their structure and characteristics. Organisms are named by the binomial system of genus and species. As evidence of internal structures became more developed due to improvements in microscopes and progress with the understanding of biochemical processes, new models of classification have been proposed. Modern classifications systems are based on theories about evolution developed from analysis of differences in DNA molecules.	WS 1.1 Show how new methods of investigation and new discoveries led to new scientific ideas.

4.4.4.5 Selective breeding

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain the impact of the selective breeding of food plants and domesticated animals.	Selective breeding (artificial selection) is the process by which humans breed plants and animals for particular genetic traits. Selective breeding involves choosing parents from a mixed population with the desired characteristic. They are bred together. From the offspring those with the desired characteristic are bred together. This continues over many generations until all the offspring show the desired characteristic. The trait can be chosen for usefulness or appearance. Selective breeding can lead to 'inbreeding' where some breeds are particularly prone to disease or inherited defects.	WS 1.3, 1.4 Evaluate the benefits and risks of selective breeding given appropriate information and consider related ethical issues.

4.4.4.6 Genetic engineering

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe genetic engineering as a process which involves modifying the genome of an organism to introduce desirable characteristics.	In genetic engineering, selected genes from one organism are transferred to another organism which may, or may not, belong to the same species. This process for genetic modification uses enzymes and vectors (such as bacterial plasmids or viruses) to transfer genes. It is much faster than selective breeding.	WS 1.4 Evaluate the advantages and disadvantages of GM technologies based on data or other information.
(HT only) Describe the main steps in the process of genetic engineering.	Genes can be transferred to the cells of animals, plants or microorganisms at an early stage in their development so that they develop with the desired characteristics.	WS 1.3 Give a simple ethical argument about the rights and wrongs of a GM technology.
Explain some of the possible benefits and risks, including practical and ethical considerations, of using gene technology in modern agriculture.	Crops that have had their genes modified in this way are called genetically modified crops (GM crops). Crops can be genetically modified to give increased yields or to increase the amount of a vitamin in the food from the crop. Genetically modified crops also include ones that are resistant to insect attack or to herbicides. This means that farmers can cut down on the use of pesticides. They can also spray to kill weeds while leaving the crop plant unaffected. Concerns about GM crops include the effect on populations of wild flowers and insects as a result of cross-pollination. Insects may evolve to become resistant so that the GM crops are no longer protected.	Recognise, in given information, the difference between a practical and an ethical argument.